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Ivri

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(54) **HYDRODYNAMICALLY ACTUATED
PRESERVATIVE FREE DISPENSING
SYSTEM**

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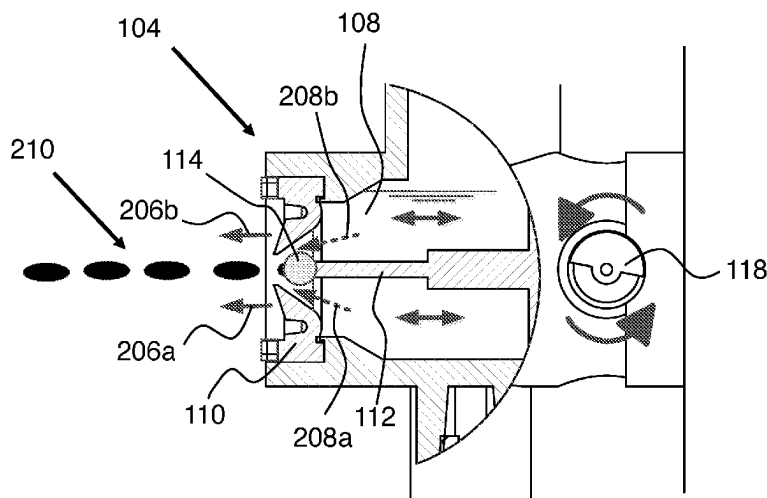
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ABSTRACT

Multi-dose preservative-free ocular fluid delivery devices
are provided. The fluid delivery device includes a fluid
dispensing system and a fluid package for storing a liquid
therein and supplying said liquid to the dispensing system.
The dispensing system comprises an elongated chamber
which includes a check valve which defines a frontal closure
to the chamber. The valve is normally closed and hermeti-
cally seals the chamber. The dispenser includes a vibration
motor that induces oscillations to the chamber and to the
fluid within. The oscillations of the chamber impart momen-
tum to the fluid stored in the chamber which in turn imparts
force that cyclically opens the valve to dispense streams or
liquid droplets. Fluid is dispensed only when the motor
oscillates while otherwise the valve is hermetically closed.

15 Claims, 6 Drawing Sheets



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See application file for complete search history.

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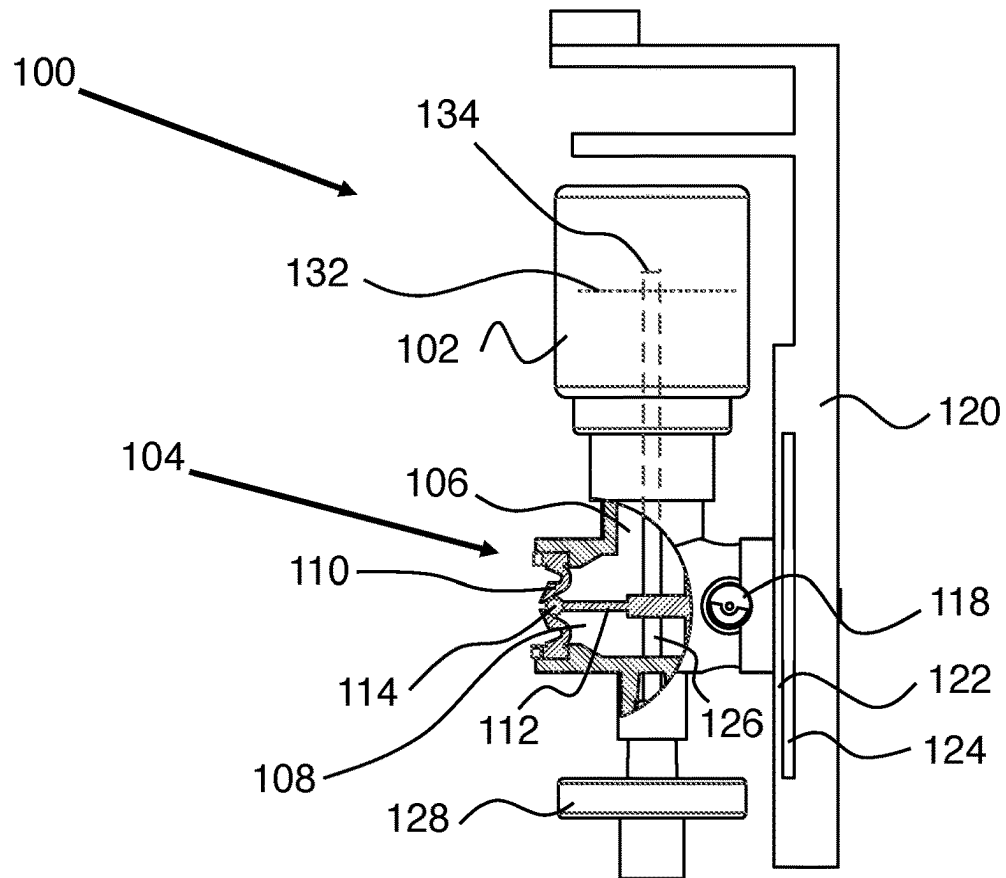


FIG. 1A

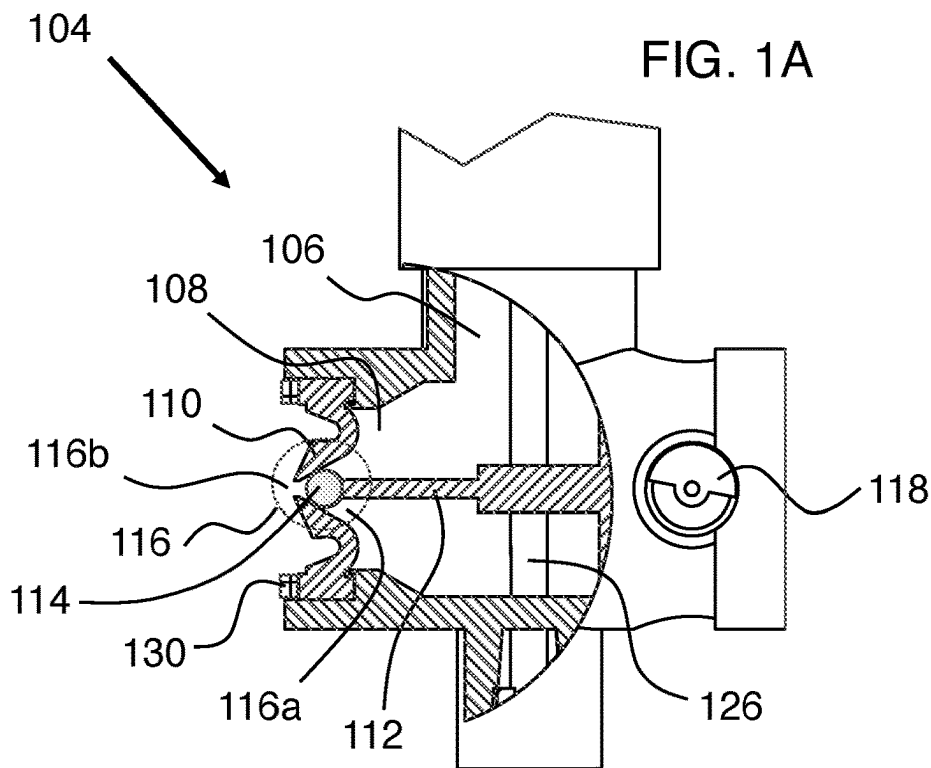


FIG. 1B

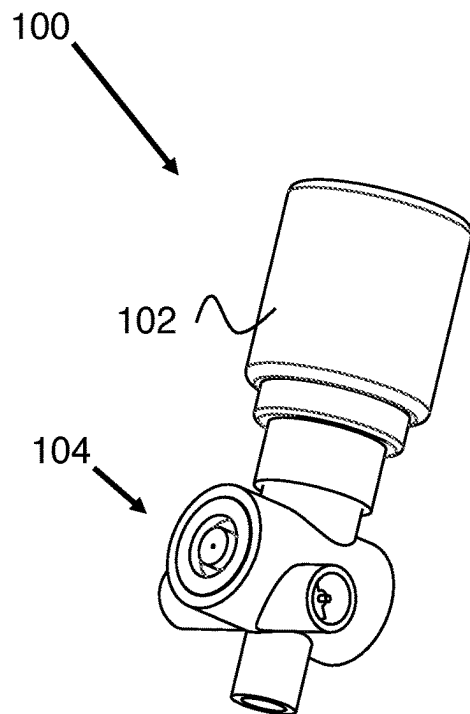


FIG. 1C

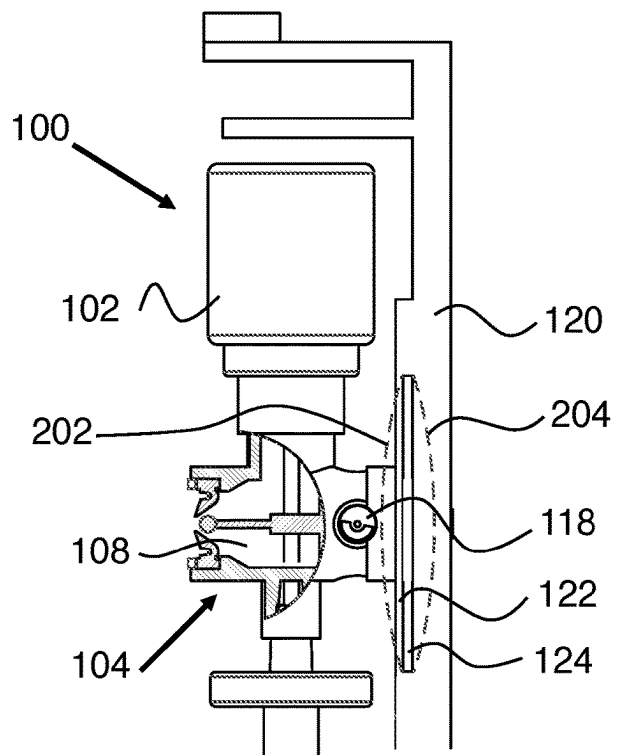


FIG. 2A

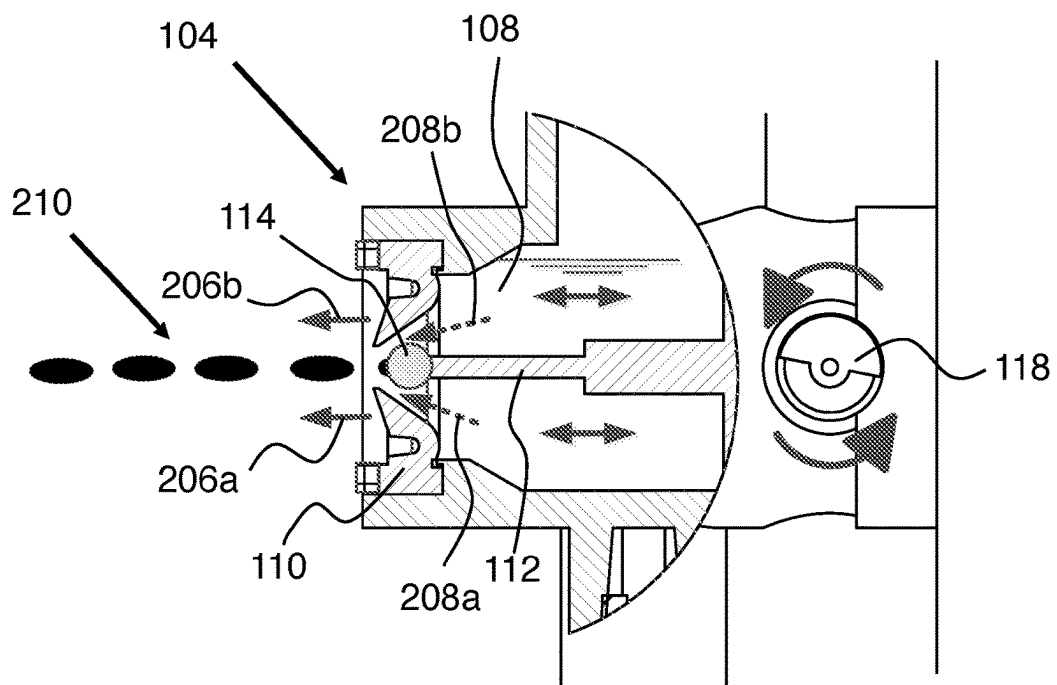


FIG. 2B

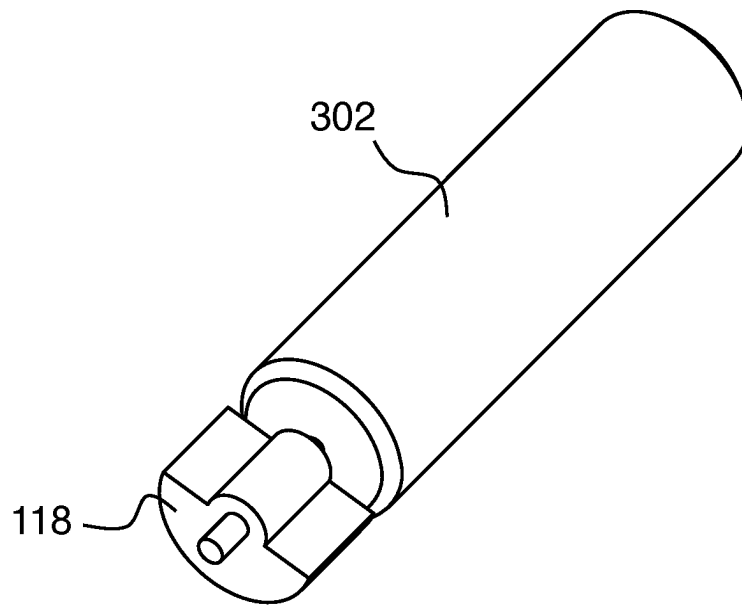


FIG. 3

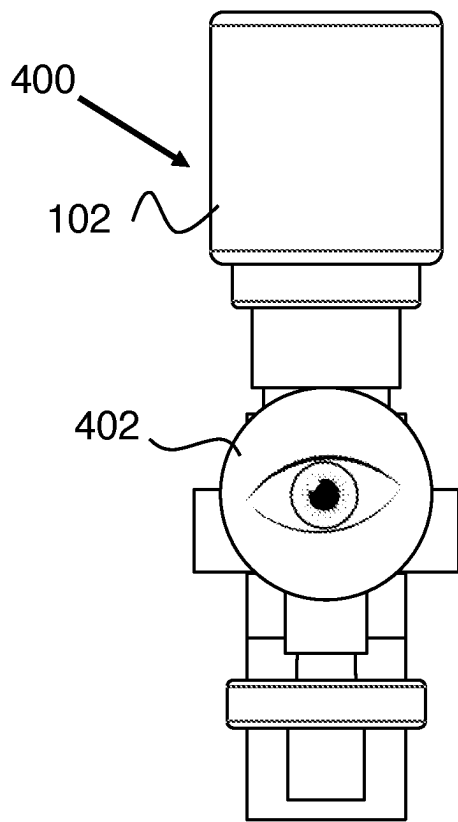


FIG. 4A

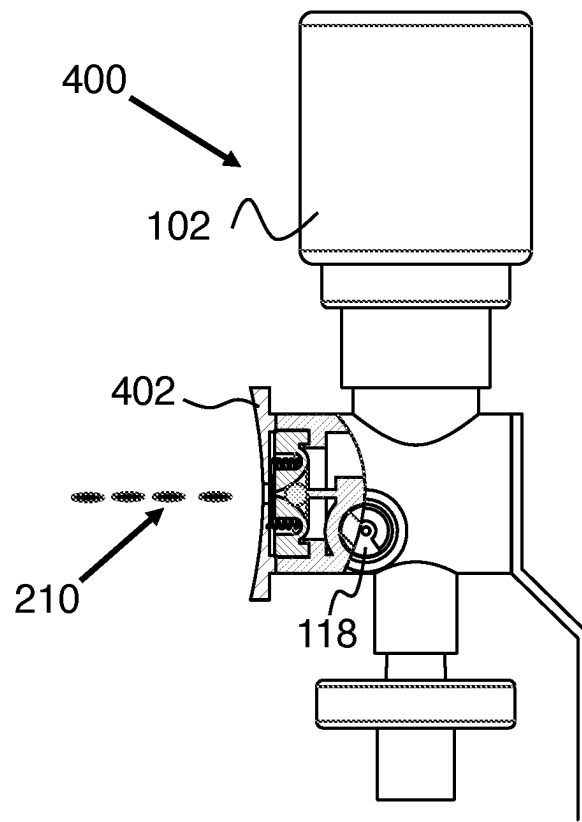


FIG. 4B

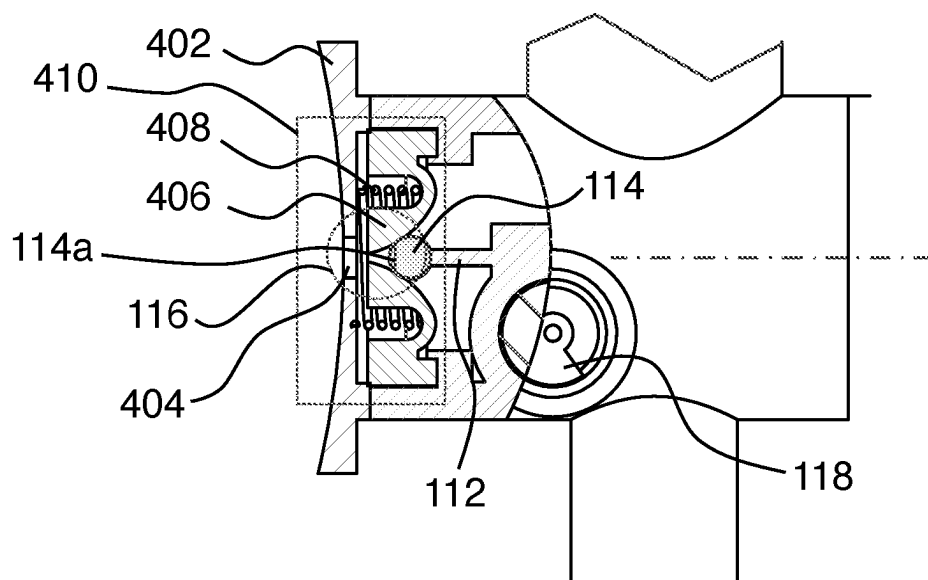


FIG. 4C

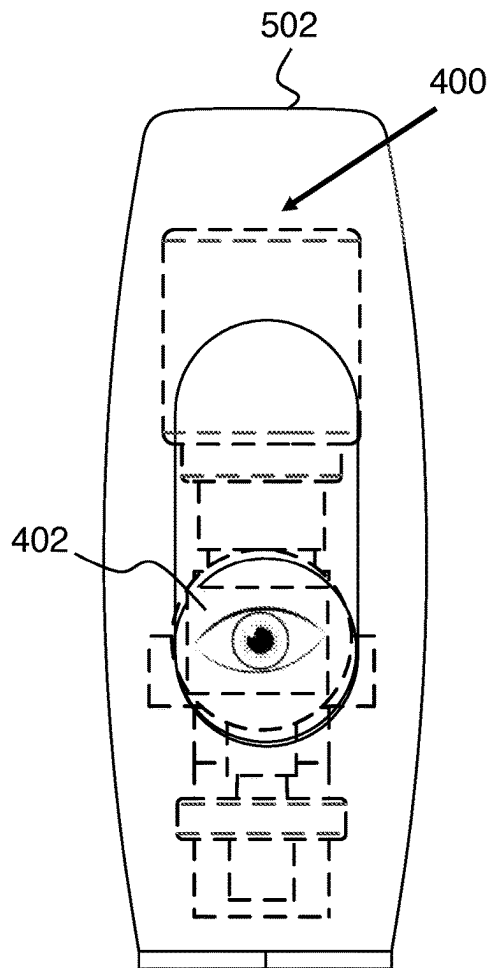


FIG. 5A

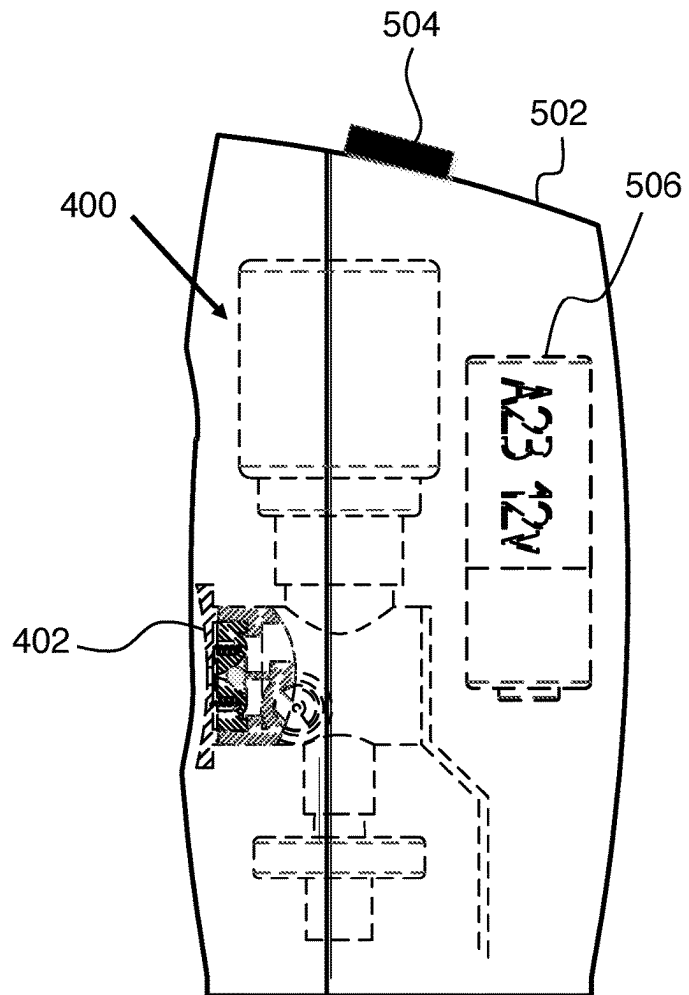


FIG. 5B

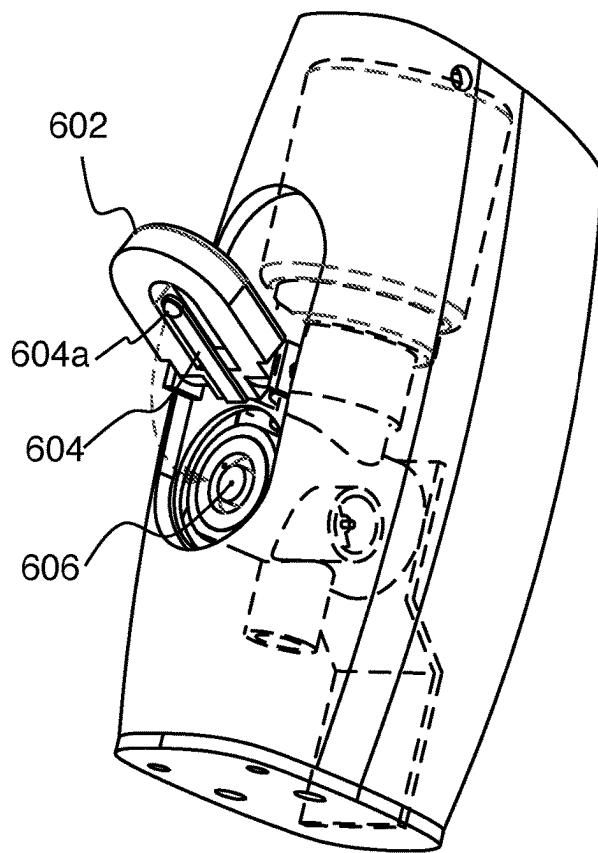


FIG. 6A

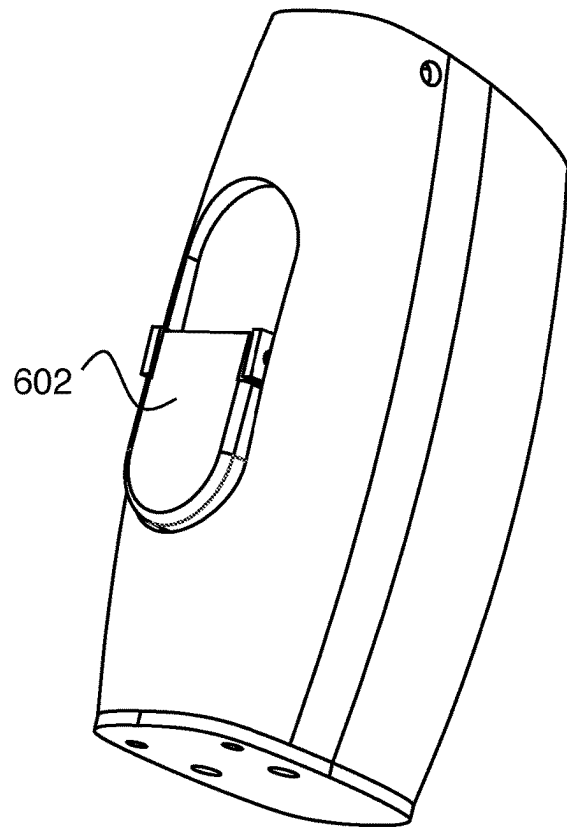


FIG. 6B

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HYDRODYNAMICALLY ACTUATED PRESERVATIVE FREE DISPENSING SYSTEM

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation patent application of U.S. application Ser. No. 17/233,105, filed Apr. 16, 2021, which claims the benefit of and priority to U.S. Provisional Patent Application No. 63/011,808, filed on Apr. 17, 2020, each of which is hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

The present invention generally pertains to devices for dispensing fluid medicines and more particularly pertains to such devices that store and deliver ophthalmic preservative-free medicines, specifically configured to increase the ease of use and enhance patient compliance with dosing instructions for the medicine.

BACKGROUND

Ease of dispensing fluid medicines and compliance with dosing instructions are primary concerns with all patients. In particular, preservative-free dispensing bottles such as ophthalmic squeeze dispensers typically require greater actuation force due to a valve mechanism that seals the dispensing nozzle to prevent bacterial ingress and contamination. Such system requires much higher pressure to operate and hence much higher squeeze force is required. In addition, prior art dispensing bottles dispense only in an upside-down orientation which require an inconvenient head maneuver and which together with higher actuation force further increases the inconvenience.

Dispensers of the kind in question are known from the prior art, for example from U.S. Pat. Nos. 6,095,376, 9,676, 525, US 2014/0336596, US 2016/0107180, U.S. Pat. Nos. 9,238,532, 8,056,766, 8,863,998, and 10,105,720. The dispenser shown in US 2014/0336596 comprises an outlet channel which connects the liquid reservoir to the outlet opening through an outlet valve which is arranged in the outlet channel and which opens when the bottle is squeezed and pressure is generated. Such preservative-free squeeze bottles typically require about 25-28N of squeeze force (*Ophthalmic Squeeze Dispenser—Drug Development and Delivery* October 2017 Vol. 17 No. 7 page 40). Elderly patients, or other patients lacking enough strength and/or dexterity in their hands, often experience problems dispensing medicine from such bottles.

This work provides preservative free ocular dispensing device that can be held horizontally, or in any convenient orientation while the actuation is done effortlessly by an electrical switch. This provides a cost effective solution that is consistent with standard drug packaging processes.

SUMMARY

Multi-dose preservative-free ocular fluid delivery devices are provided. The fluid delivery device includes a fluid dispensing system and a fluid package for storing a liquid therein and supplying said liquid to the dispensing system. The dispensing system comprises an elongated chamber which includes a check valve which defines a frontal closure to the chamber. The valve is normally closed and hermeti-

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cally seals the chamber. In this work the chamber includes a vibration motor that induces oscillations to the chamber and to the fluid within. The oscillations of the chamber impart momentum to the fluid stored in the chamber which in turn imparts force that cyclically opens the valve to dispense streams or liquid droplets. Fluid is dispensed only when the motor oscillates while otherwise the valve is hermetically closed.

The check valve can include a flexible plate which includes a conical aperture that extends through its thickness, the valve can further include a stationary spherical member that engages tangentially with the inner wall of the conical aperture to create a hermetic sealed closure. The plate can be made of elastomer that has a modulus of elasticity ranging between 0.1-1.2 GPa. The circumference of the plate can be attached to the chamber by a retaining ring that engages with the chamber in an interference fit to create the hermetically sealed closure.

The conical aperture extends through the thickness of the plate such that droplets are dispensed through the smaller opening of the aperture while the larger side of the aperture is in fluid communication with the chamber.

The spherical member may include an antibacterial coating which covers the area of the spherical member that is between the tangential engagement line and the small opening of the aperture.

The vibrational motor oscillates the chamber and the fluid within the chamber. Consequently, cycles of hydrodynamic pulses are generated causing the valve to cyclically open and dispense fluid. Here this phenomenon is characterized by oscillatory interactions between the valve and the surrounding fluid. The hydrodynamic force generated by the momentum of the fluid opens the valve and allows fluid flow through the aperture.

Fluid is dispensed only when the hydrodynamic force is sufficiently high to deform the aperture while otherwise the aperture hermetically seals the chamber. The system prevents ingress of microorganism into the chamber allowing storage of preservative free pharmaceutical. This work provides an electrically operated preservative-free dispensing system that is convenient and cost effective.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a cross sectional view of an embodiment of the invention.

FIG. 1B is a detail cross sectional view of the embodiment of FIG. 1A.

FIG. 1C is a 3D view of the embodiment of FIGS. 1A-B.

FIG. 2A is a side view and partial cross section view of an embodiment of the invention in operation.

FIG. 2B is a detail cross section view of the example of FIG. 2A.

FIG. 3 illustrates a perspective view of a vibration motor as used in embodiments of the invention.

FIG. 4A is a frontal view of an embodiment of the invention.

FIG. 4B is a side view of the embodiment of FIG. 4A.

FIG. 4C is a detail cross sectional view of the embodiment of FIG. 4A.

FIG. 5A is a frontal view of an exemplary device enclosed in a housing.

FIG. 5B is a side view of the example of FIG. 5A.

FIGS. 6A-B are perspective views of a device having a housing that includes a swivel cover.

DETAILED DESCRIPTION

This work describes dispensing devices and methods for delivery of preservative-free solutions or suspensions for

ocular administration of ophthalmic drugs. The dispensing devices include a droplet ejecting system that is fluidly connected to an ampoule package containing a liquid to be dispensed. The droplet ejecting system includes a chamber having a check valve that defines a front closure to the chamber. The dispensing system further includes a vibration motor that oscillates the chamber and induces hydrodynamic pulses which consequently causes the valve to cyclically open and eject fluid droplets. The valve is normally closed and hermetically sealing the chamber. The valve opens exclusively in response to hydrodynamic pulses induced by the oscillation of the chamber. In this way fluid is dispensed only when the device is actuated while otherwise the aperture hermetically seals the device and prevents ingress of bacterial and microorganisms thereby allowing storage of preservative-free pharmaceutical formulation. The use of a vibration motor further enables convenient and cost effective, electronically controlled administration.

FIG. 1A and FIG. 1B illustrate a side view and an enlarged partial section view, respectively, of a fluid delivery device **100**. Delivery device **100** includes a fluid reservoir **102** and dispensing system **104** connected to each other in fluid transmission relationship through passage **106**. Dispensing system **104** includes a fluid chamber **108** and also includes a check valve including aperture plate **110** which provides frontal closure to the chamber **108**. Referring to FIG. 1B it can be seen that aperture plate **110** includes a conical or tapered aperture **116** (shown inside a dotted circle for clarity) that extends through its center thickness and has a large inlet opening **116a** in fluid communication with chamber **108** and a smaller exit opening **116b** through which fluid droplets will be dispensed. Aperture plate **110** can be made of a flexible elastomer such as silicone rubber, VersaFlex, manufactured by VersaFlex Incorporated Kansas City, Kansas USA. Other elastomers with Young modulus of elasticity value between 0.5 GPa to 2 GPa can also be used. Dispensing system **104** further includes a stationary spherical member **114** that tangentially engages in pressure transmission relationship with the inlet opening **116a** of the conical aperture providing hermetically sealed closure. In a preferred embodiment spherical member **114** is made of high density polyethylene (HDPE) that is harder than silicone thereby creating a tight closure as it engages with the softer aperture plate **110**. Spherical member **114** preferably engages in pressure transmission relationship with the conical aperture **116a** with a preload force of between 0.01N and 0.05N. The combination of aperture plate **110** and spherical member **114** provides a check valve as described above. Spherical member **114** is supported by pin member **112**. Member **114** can have a shape other than spherical, since any shape capable of forming a good seal with aperture plate **110** can be used.

Aperture plate **110** can be retained to dispensing system **104** by a retaining ring **130**, thereby creating a hermetically sealed closure.

Dispensing system **100** includes a venting tube **126** that is configured to equalize the pressure inside container **102** as the fluid is dispensed from the device. The opening **134** of venting tube **126** is extended above the fluid level **132** at any orientation that the device is held. Vent tube **126** can be connected via a 0.22 micron filter **128** to assure that the air that enters the device is sterile.

Dispensing system **100** further includes a vibration motor configured to oscillate chamber **108** and the fluid within the chamber. Here this motor is schematically shown as eccentric mechanical load **118** which vibrates the assembly as

described when it is rotated by the motor (motor not shown in FIGS. 1A-B, but described below in connection with FIG. 3).

FIG. 1C is a 3D view of the embodiment of FIGS. 1A-B.

FIGS. 2A-B illustrate the response of the dispensing system **104** to oscillations. Vibrations of the motor induce oscillation to the body of dispensing system **104** which in turn generates fluid momentum within the chamber due to the dynamical interaction of the fluid with the solid structure of chamber **108**. This phenomenon is often referred to as Solid-Fluid-Interaction (SFI). The momentum of the moving fluid exerts force that flexes the aperture plate **110** outwardly in the direction indicated by arrows **206a** and **206b** causing the aperture plate **110** to disengage from spherical member **114** thereby opening a fluid flow passage as indicated by the arrows **208a** and **208b** and ejection of fluid droplets **210**.

Dispensing device **100** is supported by a flexible beam **122** or other structural embodiments which allows it to oscillate freely, as schematically shown by motion excursions **202** and **204**. Preferably the spring constant of the beam **122** is 0.05 N/mm to 0.5 N/mm. For example, beam **122** can be formed by fabricating a slot **124** in support structure **120** such that the resulting beam **122** has a thickness suitable for providing a spring constant as recited above.

FIG. 3 illustrates an exemplary vibrational motor **302**. The DC Motor **302** of this example has a cylindrical body with a diameter of 4 mm and total length of 17 mm. An eccentric flywheel **118** is attached to the motor shaft. The flywheel has a mass of 1.7 grams with a center of mass about 0.7 mm from the center of rotation. The motor receives 4.5-12 VDC and rotates at 6000-12000 RPM generating centrifugal force of 0.3N at a rotation speed of 8000 RPM. Other DC motors that generate centrifugal force of 0.1-1N and rotation speed of 1000-50000 RPM can be used. The motor can be controlled by a timer circuit which is set to provide an ON time as required to deliver a dose of 8-12 micro-liter. The actuation ON time is 60-200 ms depending on the rheology of the fluid in use. A timer circuit which incorporates a 555 timer IC or a microprocessor-based timer with a 12 volt battery such as A23 alkaline battery may be used.

FIGS. 4A-C illustrate an alternative preferred embodiment of dispensing system **400**. FIG. 4A illustrates a frontal view and FIG. 4B illustrates a side view of dispensing system **400**. Dispensing system **400** includes a concave mirror **402** which assists in aligning device **400** and fluid stream **210** to the eye of the user. In use, dispensing device **400** is positioned in front of the eye such that the image of the eye appears sharply and in the center of mirror to the user. At this point the device is properly positioned in terms of the distance of the eye from the nozzle **404** and its angular orientation relative to the eye. Upon actuation, stream **210** will be deposited precisely on the corneal surface of the eye. Device **400** preferably includes a 0.22 micron air filter configured to filter the vented air that flows into device **400**, as in the previous example.

Device **400** includes a check valve having an aperture plate **406** with a conical aperture **116** that extends through its thickness. The check valve further includes a spherical member **114** that tangentially engages with the inlet opening of the conical aperture **116**. In this example, the check valve also includes a compression spring **408** configured to force aperture plate **406** against spherical member **114**. In this way a tight seal is created along the engagement line **114a**, thus creating a tight and hermetic closure.

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Spherical member **114** can be partially covered with an antimicrobial coating, specifically in the area of spherical member **114** that is not in contact with fluid in the chamber. The coated area thus extends between the engagement line **114a** and the outlet of the conical aperture (i.e., to the left of **114a** on FIG. 4C)). Examples of antimicrobial coatings include silver in metallic form, silver alloy or a non-metallic material that contains silver including salts of silver chloride and silver sulfadiazine. Other optional material includes biguanide derivatives, chlorohexidine diacetate and chlorohexidine digluconate.

FIGS. 5A-B illustrate frontal and side views, respectively, of a dispensing device **400** packaged in a housing **502**. Housing **502** provides a convenient enclosure to the dispensing system **400** that was described in relation in FIGS. 4A-C and is shown with dashed lines here. Housing **502** includes an electrical circuit (not shown) and a 12 volt battery (e.g., type A23) **506**. The circuit controls the dispensing period such that a dose of 8-12 microliter is delivered to the ocular surface of the eye. Momentary switch **504** can be used to activate the device **400** such that a stream of droplets is ejected from the aperture as described earlier. FIGS. 5A-B also illustrate the mirror **402** which is visible on the front side of the device. Such a housing can also be used for a dispensing device **100** as described above.

FIGS. 6A-B illustrate some preferred features of housings that can be used with embodiments such as device **100** and device **400** as described above. In this example, the housing includes a swivel cover **602** that covers dispensing nozzle **606** during periods of non-use. Swivel cover **602** provides a means to prevent bacterial contamination on the external areas where a residual fluid may be left following each use. Such residual fluid may contaminate subsequent stream as it is dispensed through the nozzle. Swivel cover **602** preferably includes a flexible member **604** that includes a surface **604a** that is covered with antimicrobial coating. Surface **604a** engages with the outlet opening of nozzle **606** when the swivel cover **602** is closed as illustrated in FIG. 6B. In this way antipathogen action is achieved. Alternatively, organic dyes with antiseptic action may also be used. For example, toluidine blue, methylene blue, gentian violet and acridine and related active substances such as acridine orange and acridine yellow as well as ethacridine lactate. Germicidal polymers such as polyhexanide are also possible. Materials that contain additives which contain metal-organic substances with an ionizing effect can also be used. Such additives are available from SteriOne GmbH Berlin, Germany. Examples of antimicrobial coatings that can be used on surface **604a** include silver in metallic form, silver alloy or a non-metallic material that contains silver including salt of silver chloride and silver sulfadiazine. Other optional material includes biguanide derivatives, chlorohexidine diacetate and chlorohexidine digluconate.

The invention claimed is:

1. A method for delivering fluid to an eye of a patient, the method comprising:
 using an actuator to oscillate a chamber so as to generate fluid momentum from the fluid therein;
 allowing the fluid momentum to disengage an aperture from a pin member disposed within the chamber to eject the fluid through the aperture; and

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reengaging the aperture with the pin member to seal the aperture after the fluid is delivered.

2. The method of claim 1, wherein the aperture and the pin member are hermetically sealed when the fluid is not being ejected.

3. The method of claim 1, wherein the aperture is a flexible aperture.

4. The method of claim 1, further comprising:
 biasing the aperture against the pin member via a compression spring.

5. The method of claim 1, wherein the reengaging step further comprises preventing an ingress of bacteria or contaminants as a result of the engagement between the aperture and the pin member.

6. The method of claim 1, further comprising:
 providing a mirror such that when the eye of the patient is in-focus and centered in the mirror, ejection of the fluid through the aperture will be delivered accurately to the eye.

7. The method of claim 1, further comprising:
 equalizing a pressure of the fluid as it is ejected through the aperture.

8. The method of claim 7, wherein the equalizing step includes a vent in fluid communication with the chamber and ambient air and introduces air into the chamber as the fluid is ejected.

9. The method of claim 8 further comprising:
 filtering the air to ensure the air is sterile before introduction into the chamber.

10. An apparatus for delivering a fluid to an eye of a patient, the apparatus comprising:

- a) a reservoir to accommodate the fluid;
- b) a liquid dispenser in fluid communication with the reservoir, the liquid dispenser having a check valve through which the fluid can be selectively dispensed;
- c) an actuator positioned to vibrate the liquid dispenser and generate fluid momentum from the fluid therein to open the check valve and eject the fluid therethrough; and

wherein the check valve is a pin member engaging an aperture.

11. The apparatus of claim 10, wherein the actuator includes a vibration motor having an eccentric mechanical load relative to an axis of rotation of the vibration motor.

12. The apparatus of claim 10, wherein the check valve includes the aperture which is tapered to be narrower at an outlet of the aperture than at an inlet of the aperture.

13. The apparatus of claim 10, further comprising a vent to equalize a pressure of the fluid as the fluid is ejected.

14. The apparatus of claim 13, wherein the vent includes a filter configured to ensure that, as ambient air is introduced, the ambient air that enters is sterile.

15. The apparatus of claim 10, further comprising a concave mirror configured to provide an in-focus image of the eye of the patient to the patient when the liquid dispenser is positioned properly to deliver the fluid to the eye of the patient.

* * * * *